

Spectral-gap decay and the ground-state asymptotic

A substantial answer to the decay-range question in arXiv:2306.02828

Automated mathematical discovery run `fa.banach.001`, lane 9

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Candidate substantial partial result. For the small-data global solution constructed in the source paper, the reported polynomial decay is never the true large-time scale. After a possibly smaller smallness threshold, the solution decays in every $L^a(\mathbb{R}^d)$, $p \leq a \leq \infty$, and in $\exp L^p$, at every exponential rate below d^β . More precisely, in L^2 it equals a multiple of the harmonic-oscillator ground state times $e^{-d^\beta t}$, up to a remainder with every rate below $\min\{(d+2)^\beta, md^\beta\}$.

Source question and transcription

The source studies

$$\partial_t u + H^\beta u = f(u), \quad H = -\Delta + |x|^2, \quad 0 < \beta \leq 1,$$

with small initial data in $\exp L^p(\mathbb{R}^d)$ and a nonlinearity satisfying

$$|f(s) - f(t)| \leq C|s - t|(|s|^{m-1}e^{\lambda|s|^p} + |t|^{m-1}e^{\lambda|t|^p}). \quad (\text{S})$$

Its Theorem 1.2 constructs a global weak-mild solution with uniform Orlicz control and a polynomial L^a -estimate for a restricted range of exponents. Remark 1.2(iii) asks, verbatim: “A natural question arises as to whether the range of a required for (1.5) is optimal.”

This packet completely determines the large-time mechanism, but does not settle the source’s separate short-time endpoint restrictions.

Original source image

Remark 1.2. In view of hypothesis and conclusions stated in the above theorem, some comments arise, we enumerate them in what follows.

- (i) The restriction on β in Theorem 1.2 is a direct consequence of condition (2) in Theorem 2.1 below.
- (ii) The assumption $p > 1$ is essential in the subsequent Corollaries 2.1 and 2.2.
- (iii) A natural question arises as to whether the range of a required for (1.5) is optimal. It is likely that this range can be improved, and further investigation could refine our understanding of the conditions under which (1.5) holds.
- (iv) Similar results have been obtained in previous studies such as [8, 26, 27] for the standard NLH equation, where $\varrho = 0$ in (1.1).

Source: Bhimani–Majdoub–Manna, arXiv:2306.02828v3, PDF p. 4, Remark 1.2. This is a direct crop from `source_paper.pdf`.

Definitions and source input

The Luxemburg norm on $\exp L^p(\mathbb{R}^d)$ is

$$\|v\|_{\exp L^p} = \inf \left\{ A > 0 : \int_{\mathbb{R}^d} \left(e^{|v(x)|^p/A^p} - 1 \right) dx \leq 1 \right\}.$$

The source fixed-point argument supplies at least one admissible exponent $a_0 > p$, numbers $M, \sigma_0 > 0$, and a global solution satisfying

$$\sup_{t>0} \|u(t)\|_{\exp L^p} \leq M, \quad \|u(t)\|_{a_0} \leq Mt^{-\sigma_0}. \quad (1)$$

The small-data threshold can be decreased so that M is as small as needed below. We use two estimates already recorded in the source:

$$\|v\|_q \leq C_{p,q} \|v\|_{\exp L^p}, \quad p \leq q < \infty, \quad (2)$$

$$\left\| e^{-tH^\beta} g \right\|_q \leq C e^{-d^\beta t} \|g\|_r, \quad t \geq 1, \quad 1 \leq r, q \leq \infty, \quad (3)$$

$$\left\| e^{-tH^\beta} g \right\|_q \leq C t^{-\frac{d}{2\beta}|1/r-1/q|} \|g\|_r, \quad 0 < t \leq 1. \quad (4)$$

Here and below, endpoint estimates are available because $0 < \beta \leq 1$.

Idea / proof intuition

The restricted exponent a_0 is only a bootstrap seed. Its polynomial decay, together with uniform Orlicz moments, makes $f(u(t))$ decay in a sufficiently high L^r . Restarting the mild equation one time unit in the past then forces $u(t) \rightarrow 0$ in L^∞ . Once this happens, (S) becomes $|f(u)| \leq \kappa \delta^{m-1} |u|$, where δ can be made arbitrarily small by moving the restart time. The spectral gap in (3) therefore dominates the nonlinearity and yields exponential decay in $L^p \cap L^\infty$.

Finally, the discrete spectrum $d^\beta, (d+2)^\beta, \dots$ separates the ground state from all higher modes. The nonlinear forcing is of order $e^{-m\omega t}$, so a scalar variation-of-constants formula gives the leading ground-state coefficient, while the orthogonal complement decays at the minimum of the next spectral rate and the nonlinear rate.

Main result

Let $\phi_0(x) = \pi^{-d/4} e^{-|x|^2/2}$, normalized in L^2 , and put

$$\lambda_0 = d^\beta, \quad \lambda_1 = (d+2)^\beta.$$

Theorem 1 (Large-time range and asymptotic profile). *Assume the hypotheses of the source's global-existence theorem: $1 < p \leq d(m-1)/(2\beta)$, $m \geq p$, $0 < \beta \leq 1$, and (S), with $f(0) = 0$. After possibly decreasing its small-data constant, the global solution constructed there has the following properties.*

(i) *For every $a \in [p, \infty]$ and every $0 < \omega < \lambda_0$, there are $T_\omega, C_{a,\omega} > 0$ such that*

$$\|u(t)\|_a + \|u(t)\|_{\exp L^p} \leq C_{a,\omega} e^{-\omega t}, \quad t \geq T_\omega. \quad (5)$$

(ii) *There is a scalar c such that, for every $0 < \mu < \min\{\lambda_1, m\lambda_0\}$,*

$$\left\| u(t) - ce^{-\lambda_0 t} \phi_0 \right\|_2 \leq C_\mu e^{-\mu t} \quad (6)$$

for all sufficiently large t .

Proof. We divide the argument into four steps.

Step 1: a decaying forcing norm. Choose

$$r > \max \left\{ \frac{d}{2\beta}, \frac{a_0}{m} \right\}.$$

By decreasing the source smallness threshold, assume that the uniform bound in (1) is small enough that $2r\lambda(2M)^p \leq 1$. The Luxemburg definition and $(e^x - 1)^{2r} \leq e^{2rx} - 1$ give

$$\sup_{t>0} \left\| e^{\lambda|u(t)|^p} - 1 \right\|_{2r} \leq C. \quad (7)$$

The Orlicz embedding (2) uniformly bounds every finite L^q -norm. Since $a_0 < mr < 2mr < 4mr$, interpolation between the decaying L^{a_0} -norm and the bounded L^{4mr} -norm gives positive numbers θ_1, θ_2 such that

$$\begin{aligned} \|u(t)\|_{mr} &\leq C(1+t)^{-\theta_1\sigma_0}, \\ \|u(t)\|_{2mr} &\leq C(1+t)^{-\theta_2\sigma_0}. \end{aligned}$$

Putting the second scalar argument equal to zero in (S) yields $|f(v)| \leq C|v|^m e^{\lambda|v|^p}$. Hence, by (7),

$$\|f(u(t))\|_r \leq C \|u(t)\|_{mr}^m + C \|u(t)\|_{2mr}^m \left\| e^{\lambda|u(t)|^p} - 1 \right\|_{2r} \leq C(1+t)^{-\eta} \quad (8)$$

for some $\eta > 0$.

Step 2: entry into a small L^∞ -ball. For $t \geq 2$, restart the mild equation at $t - 1$:

$$u(t) = e^{-H^\beta} u(t-1) + \int_{t-1}^t e^{-(t-s)H^\beta} f(u(s)) ds. \quad (9)$$

The fixed-time L^{a_0} -to- L^∞ estimate and (1) make the first term tend to zero. For the integral, (4), (8), and $\gamma = d/(2\beta r) < 1$ give

$$\left\| \int_{t-1}^t e^{-(t-s)H^\beta} f(u(s)) ds \right\|_\infty \leq C(1+t)^{-\eta} \int_0^1 \tau^{-\gamma} d\tau \longrightarrow 0. \quad (10)$$

Thus $\|u(t)\|_\infty \rightarrow 0$.

Step 3: spectral-gap bootstrap. Fix $0 < \omega < \lambda_0$ and choose $\omega < \nu < \lambda_0$. Combining the short- and long-time parts of (3)–(4), for $q = p, \infty$ there is A such that

$$\left\| e^{-tH^\beta} \right\|_{L^q \rightarrow L^q} \leq A e^{-\nu t}, \quad t \geq 0. \quad (11)$$

Choose $\delta > 0$ so small that $A C e^{\lambda\delta^p} \delta^{m-1} < \nu - \omega$, and then choose T so that $\|u(t)\|_\infty \leq \delta$ for $t \geq T$. Condition (S) gives pointwise

$$|f(u(t, x))| \leq C e^{\lambda\delta^p} \delta^{m-1} |u(t, x)|.$$

Taking the sum norm on $B = L^p \cap L^\infty$, (9) restarted at T and (11) imply

$$\|u(t)\|_B \leq A e^{-\nu(t-T)} \|u(T)\|_B + A \kappa_\delta \int_T^t e^{-\nu(t-s)} \|u(s)\|_B ds,$$

where $\kappa_\delta = C e^{\lambda\delta^p} \delta^{m-1}$. Weighted Gronwall therefore gives

$$\|u(t)\|_p + \|u(t)\|_\infty \leq C_\omega e^{-\omega t}. \quad (12)$$

Interpolation proves (5) for every $p \leq a \leq \infty$. The source inclusion $L^p \cap L^\infty \hookrightarrow \exp L^p$, with $\|v\|_{\exp L^p} \leq C(\|v\|_p + \|v\|_\infty)$, proves the Orlicz part of (5).

Step 4: the ground-state coefficient. At a sufficiently large positive time T_2 , the solution belongs to L^2 . Indeed, choose T_2 so that the already proved L^∞ -smallness holds on $[T_2 - 1, T_2]$. In the one-unit restart formula the linear term is in L^2 by fixed-time smoothing from L^p . On the last unit interval, (S) and boundedness give $|f(u)| \leq C|u|^m$, while interpolation between L^p and L^∞ gives $u \in L^{2m}$; hence the forcing is in L^2 . Thus (9) is an L^2 identity from T_2 onward.

Let $P_0 v = \langle v, \phi_0 \rangle \phi_0$ and $Q = I - P_0$. Fix $\mu < \min\{\lambda_1, m\lambda_0\}$, and choose $\alpha < \lambda_0$ so close to λ_0 that $m\alpha > \max\{\mu, \lambda_0\}$. By (12), interpolation, and $2m \geq p$,

$$\|f(u(t))\|_2 \leq C \|u(t)\|_{2m}^m \leq C e^{-m\alpha t}. \quad (13)$$

The scalar ground-state equation gives

$$e^{\lambda_0 t} \langle u(t), \phi_0 \rangle = e^{\lambda_0 T_2} \langle u(T_2), \phi_0 \rangle + \int_{T_2}^t e^{\lambda_0 s} \langle f(u(s)), \phi_0 \rangle ds.$$

The integral converges by (13). Define

$$c = e^{\lambda_0 T_2} \langle u(T_2), \phi_0 \rangle + \int_{T_2}^\infty e^{\lambda_0 s} \langle f(u(s)), \phi_0 \rangle ds. \quad (14)$$

The omitted tail is $O(e^{-m\alpha t})$ after restoring the factor $e^{-\lambda_0 t}$. On the orthogonal complement, the spectral theorem and (13) yield

$$\|Qu(t)\|_2 \leq e^{-\lambda_1(t-T_2)} \|Qu(T_2)\|_2 + C \int_{T_2}^t e^{-\lambda_1(t-s)} e^{-m\alpha s} ds \leq C_\mu e^{-\mu t}.$$

Together with the ground-state tail, this is (6). □

Consequences and sharpness

If the coefficient c in (14) is nonzero, (6) shows that the exact large-time exponent is d^β ; no faster uniform estimate can hold for that solution. If $c = 0$, the solution decays faster, with the next possible obstruction coming from either the first excited spectral level or the order- m nonlinear forcing. Thus the source's polynomial bound is useful at short time and as a bootstrap seed, but is far from optimal at large time.

For example, when $f \equiv 0$ and $u_0 = \varepsilon \phi_0$, one has exactly $u(t) = \varepsilon e^{-d^\beta t} \phi_0$, so the spectral rate is attained within the source hypotheses.

Verification, novelty, and scope

- Every integrability exponent used in Step 1 is finite and larger than the decaying seed exponent; the uniform high moments come directly from the Orlicz bound.
- The near-time singularity in Step 2 is integrable precisely because $r > d/(2\beta)$.
- The perturbative constant in Step 3 tends to zero with $\|u(t)\|_\infty$; this is why every rate below, rather than merely one unspecified rate below, d^β is obtained.
- The L^2 asymptotic does not assume $u_0 \in L^2$; fixed-time harmonic-oscillator smoothing supplies L^2 membership.
- Formula (14) explicitly identifies the leading coefficient and the remainder estimate separates the next spectral and nonlinear rates.

Novelty audit. Cheap run indexes were searched for arXiv:2306.02828 and the terms “optimal range”, “spectral gap”, “ground-state asymptotic”, and “Orlicz exponential nonlinearity”. Bounded arXiv/primary-source searches through 17 August 2026 found work on exponential nonlinearities and recent sharp decay estimates for other oscillator equations, but no source giving this all- a large-time theorem or the coefficient formula (14) in the present Orlicz setting. Novelty confidence is moderate, not definitive.

Limitations. The theorem answers the large-time part of the source question and gives the precise asymptotic scale. It does not enlarge the source’s admissible range in its exact all-time estimate (1.5), whose power of t also encodes the singular behavior as $t \downarrow 0$. It applies to the small-data solution constructed by the source after a possibly smaller threshold, not to arbitrary large-data solutions. The coefficient c may vanish for special data.

Human review. Pending. Priority checks are the passage from the source weak-mild representative to the one-unit restart identity in Lebesgue spaces, and the uniform exponential-moment choice after shrinking the source smallness constant.

References

- [1] D. G. Bhimani, M. Majdoub, and R. Manna, *Heat equations associated to harmonic oscillator with exponential nonlinearity*, arXiv:2306.02828v3, especially Theorem 1.2, Remark 1.2, and Theorem 2.1.
- [2] D. G. Bhimani, R. Manna, F. Nicola, S. Thangavelu, and S. I. Trapasso, *On heat equations associated with fractional harmonic oscillators*, *Fract. Calc. Appl. Anal.* 26 (2023), 2470–2492.
- [3] M. Majdoub and S. Tayachi, *Global existence and decay estimates for the heat equation with exponential nonlinearity*, *Funkcial. Ekvac.* 64 (2021), 237–259.